



ÉCOLE POLYTECHNIQUE DE SOUSSE

WINDOWS SERVER 2019 ADMINISTRATION

Active Directory Users and Groups Management

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Table des matières

1	Active Directory Users and Groups Management	3
1.1	Objective	3
1.2	Part A – Organizational Unit (OU) Design	3
1.2.1	A.1 – Role and Purpose of OUs	3
1.2.2	A.2 – Manual Creation via GUI (RSAT)	3
1.2.3	A.3 – Automated Creation via PowerShell	4
1.3	Part B – User Account Management	5
1.3.1	B.1 – Account Organization	5
1.3.2	B.2 – Manual Creation via GUI	5
1.3.3	B.3 – Verification via PowerShell	6
1.4	Part C – Security Group Management	6
1.4.1	C.1 – Group Creation via GUI	6
1.5	Verification & Validation	7
1.6	Key Takeaways	7

Table des figures

1.1	Final Organizational Unit (OU) hierarchy in Active Directory Users and Computers	4
1.2	Bulk user list retrieved via PowerShell (Get-ADUser) after OU automation script execution	5
1.3	User account attributes and password policy verification via PowerShell . .	6
1.4	Manual security group creation and membership configuration (GRP-RH) via ADUC	6
1.5	Group membership verification via PowerShell (Get-ADGroupMember) . .	7
1.6	Security group scope, category, and OU placement validation via PowerShell (Get-ADGroup)	7

Chapitre 1

Active Directory Users and Groups Management

1.1 Objective

This lab focused on deploying and administering Active Directory Domain Services (AD DS) on Windows Server 2019, a core component of enterprise network infrastructure. The work covered :

- Designing and implementing a logical **Organizational Unit (OU)** structure to organize users and computers
- Creating **users and security groups** through both manual (GUI, via RSAT — Remote Server Administration Tools) and scripted (PowerShell) methods
- Applying **fine-grained password policies**, access controls, and evaluating the operational benefits of centralized domain administration over decentralized/local account management

1.2 Part A – Organizational Unit (OU) Design

1.2.1 A.1 – Role and Purpose of OUs

An OU is a logical container within AD DS used to organize domain objects (users, groups, computers, sub-OUs) independently of the physical network topology. Unlike security groups, OUs are not security principals and cannot be granted direct resource permissions — their primary functions are :

- Group Policy (GPO) linking : GPOs applied at the OU level cascade down to all contained objects, enabling scoped configuration (e.g., password policies, software deployment, desktop restrictions) per department.
- Delegated administration : Using the Delegation of Control Wizard, specific administrative tasks (e.g., password resets) can be assigned to junior admins at the OU level without granting full Domain Admin rights — a key least-privilege security practice.
- Logical structuring : OUs mirror organizational departments (HR, IT, Students, Teaching Staff), independent of Active Directory's underlying domain/forest architecture.

1.2.2 A.2 – Manual Creation via GUI (RSAT)

Using the Active Directory Users and Computers (ADUC) console — part of the RSAT toolset used to manage AD DS remotely or locally on the Domain Controller

— I built a multi-level OU hierarchy representing a simulated organization : HR, IT (with Security/Support sub-OUs), Teaching Staff (Permanent/Contract sub-OUs), and Students (Computer Science/Networks sub-OUs).

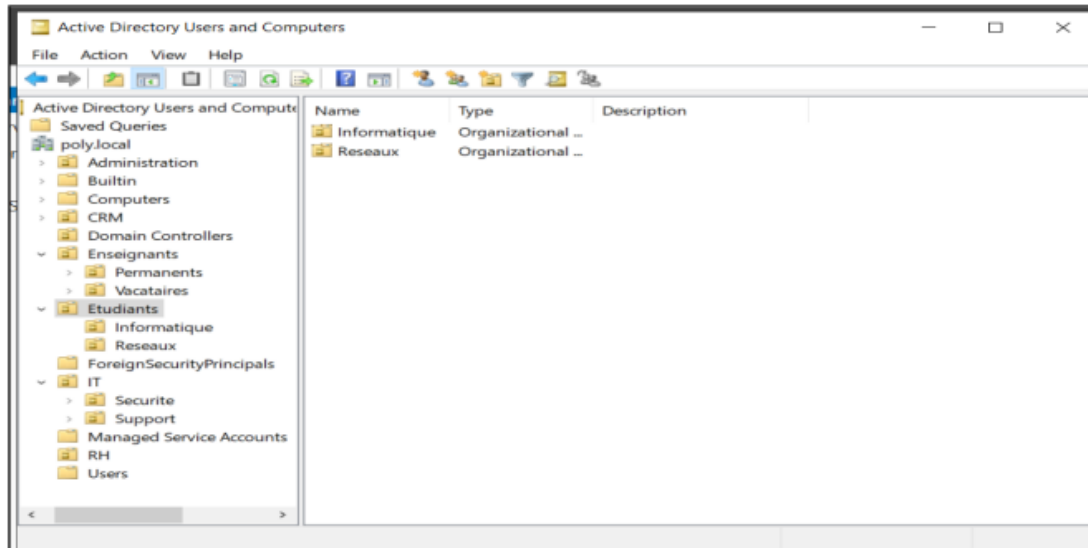


FIGURE 1.1 – Final Organizational Unit (OU) hierarchy in Active Directory Users and Computers

Each OU maps to a distinct simulated business unit, reinforcing an understanding of AD's logical hierarchy and how nested OUs allow GPOs to be inherited or blocked at different levels (GPO inheritance and enforcement).

1.2.3 A.3 – Automated Creation via PowerShell

To eliminate manual, error-prone repetition, I authored a PowerShell script (`create_OU_poly.ps1`, stored in C :

Scripts) using the **New-ADOrganizationalUnit** cmdlet to programmatically provision the full OU structure in a single execution — demonstrating Infrastructure-as-Code principles applied to directory services, a practice critical for scalable, auditable environments.

```

Windows PowerShell
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PS C:\Users\Administrator> Get-ADUser -Filter * | Select Name, SamAccountName, Enabled, DistinguishedName
>>

```

Name	SamAccountName	Enabled	DistinguishedName
Administrator	Administrator	True	CN=Administrator,CN=Users,DC=poly,DC=local
Guest	Guest	False	CN=Guest,CN=Users,DC=poly,DC=local
krbtgt	krbtgt	False	CN=krbtgt,CN=Users,DC=poly,DC=local
Mohamed ma. Ali	ma	True	CN=Mohamed ma. Ali,OU=Informatique,OU=Etudiants,DC=poly,DC=local
Sara sa. Youssef	sa	True	CN=Sara sa. Youssef,OU=Informatique,OU=Etudiants,DC=poly,DC=local
Khaled kbs. Ben Salem	kbs	True	CN=Khaled kbs. Ben Salem,OU=RH,DC=poly,DC=local
Leila la. Abdallah	la	True	CN=Leila la. Abdallah,OU=RH,DC=poly,DC=local
Fatma fz. Zahra	fz	True	CN=Fatma fz. Zahra,OU=Reseaux,OU=Etudiants,DC=poly,DC=local
Ali ah. Hassani	ah	True	CN=Ali ah. Hassani,OU=Reseaux,OU=Etudiants,DC=poly,DC=local
Amina am. Mansour	am	True	CN=Amina am. Mansour,OU=Permanents,OU=Enseignants,DC=poly,DC=local
Sami sm. Marzouk	sm	True	CN=Sami sm. Marzouk,OU=Permanents,OU=Enseignants,DC=poly,DC=local
Rim rbi. Ben Issa	rbi	True	CN=Rim rbi. Ben Issa,OU=Vacataires,OU=Enseignants,DC=poly,DC=local
Youssef yh. Hassen	yh	True	CN=Youssef yh. Hassen,OU=Vacataires,OU=Enseignants,DC=poly,DC=local
Housseem ht. Taher	ht	False	CN=Housseem ht. Taher,OU=Support,OU=IT,DC=poly,DC=local
Marouane ms. Saad	ms	False	CN=Marouane ms. Saad,OU=Support,OU=IT,DC=poly,DC=local
Walid wj. Jomaa	wj	False	CN=Walid wj. Jomaa,OU=Securite,OU=IT,DC=poly,DC=local
Anis ah. Hammadi	Aha	False	CN=Anis ah. Hammadi,OU=Securite,OU=IT,DC=poly,DC=local
Nadia ns. Nadia Souissi	ns	True	CN=Nadia ns. Nadia Souissi,OU=Administration,DC=poly,DC=local
Ahmed al. Latif	al	True	CN=Ahmed al. Latif,OU=Administration,DC=poly,DC=local

FIGURE 1.2 – Bulk user list retrieved via PowerShell (Get-ADUser) after OU automation script execution

1.3 Part B – User Account Management

1.3.1 B.1 – Account Organization

Each OU was populated with representative user accounts mapped to its department (e.g., HR, Networks/Students, IT/Security), simulating a realistic multi-department directory structure with distinct **SamAccountName** and **UserPrincipalName (UPN)** conventions.

1.3.2 B.2 – Manual Creation via GUI

Accounts were provisioned via **ADUC** → **OU** → **New** → **User**, following a standardized `firstname.lastname@poly.local` UPN convention, with passwords enforced against the domain's Default Domain Policy complexity requirements (minimum length, mixed case, numeric/special characters).

Role-based account attributes were configured to reflect real-world identity lifecycle and security practices :

- Students → User must change password at next logon (onboarding hygiene)
- HR / Administration → Password never expires (service-account-style privileged continuity)
- Teaching Staff → User cannot change password (centrally managed credentials)
- IT/Security accounts → disabled by default, activated only upon formal provisioning approval — a common zero-standing-privilege onboarding control

1.3.3 B.3 – Verification via PowerShell

Account attributes were queried and validated using `Get-ADUser -Properties *`, confirming correct **PasswordLastSet** timestamps, distinguished names (DN), and organizational placement — useful for auditing credential hygiene across the domain.

```
PS C:\Users\Administrator> Get-ADUser -Identity sm -Properties * | Select Name,PasswordLastSet,DistinguishedName
Name                PasswordLastSet      DistinguishedName
-----
Sami sm. Marzouk 10/19/2025 6:07:35 AM CN=Sami sm. Marzouk,OU=Permanents,OU=Enseignants,DC=poly,DC=local
```

FIGURE 1.3 – User account attributes and password policy verification via PowerShell

1.4 Part C – Security Group Management

1.4.1 C.1 – Group Creation via GUI

A global security group was created within each OU to implement role-based access control (RBAC), following the AGDLP/AGUDLP best-practice model (Accounts → Global groups → Domain Local groups → Permissions) used to simplify permission management at scale :

- GRP-RH
- GRP-Students-IT
- GRP-IT-Security

Security groups (vs. distribution groups) were specifically chosen since they can be assigned NTFS/share permissions and applied within GPO Security Filtering.

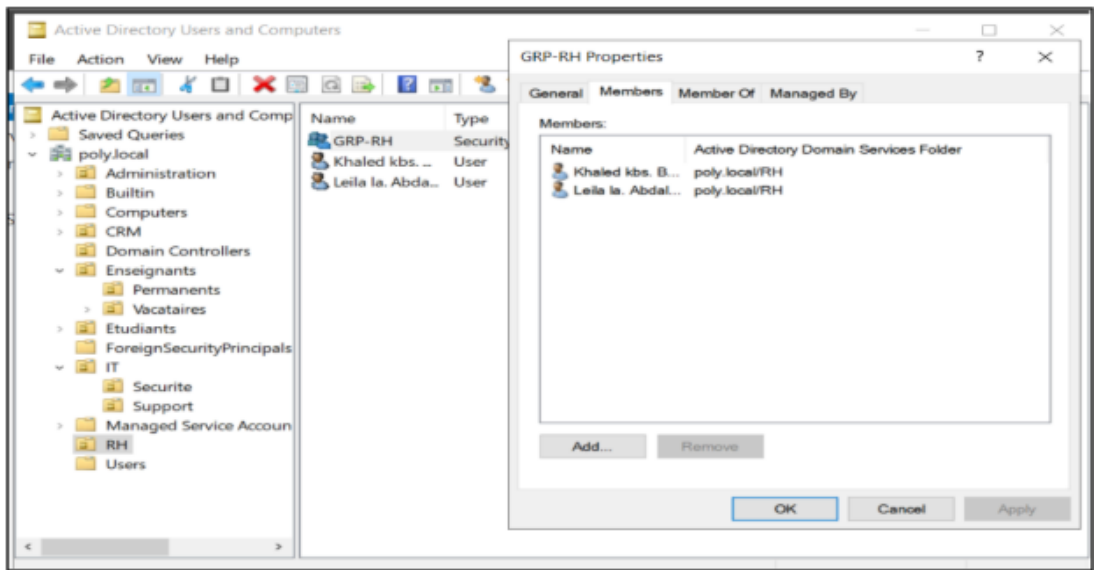


FIGURE 1.4 – Manual security group creation and membership configuration (GRP-RH) via ADUC

1.5 Verification & Validation

GUI verification : Confirmed OUs, users, and groups were correctly nested and visible in ADUC, with correct canonical names reflecting the intended hierarchy.

PowerShell verification : Used `Get-ADGroupMember` to enumerate group membership and `Get-ADGroup -Properties CanonicalName` to validate group scope (Global), category (Security), and OU placement — confirming configuration matched design intent without relying solely on GUI inspection.

```
PS C:\Users\Administrator> Get-ADGroupMember "GRP-Etudiants-Reseaux" | Select Name, SamAccountName, ObjectClass
>>

Name                SamAccountName ObjectClass
-----
Fatma fz. Zahra fz      user
Ali ah. Hassani ah      user
```

FIGURE 1.5 – Group membership verification via PowerShell (`Get-ADGroupMember`)

```
PS C:\Users\Administrator> Get-ADGroup -Filter "Name -like 'GRP-*'" -Properties CanonicalName |
>> Select Name, GroupScope, GroupCategory, CanonicalName
>>

Name                GroupScope GroupCategory CanonicalName
-----
GRP-Etudiants-Informatique Global      Security    poly.local/Etudiants/Informatique/GRP-Etudiants-Informatique
GRP-RH               Global      Security    poly.local/RH/GRP-RH
```

FIGURE 1.6 – Security group scope, category, and OU placement validation via PowerShell (`Get-ADGroup`)

1.6 Key Takeaways

This lab provided hands-on experience with **centralized identity and access management** in a Windows domain environment, covering the full lifecycle from OU design through account provisioning to group-based access control. Key comparisons observed :

- **GUI administration (RSAT/ADUC)** : intuitive and low-barrier for one-off changes, but manual, slow, and error-prone at scale
- **PowerShell automation** : more abstract initially, but faster, idempotent, and scriptable — essential for reproducible deployments, compliance auditing, and integration into larger automation/DevOps pipelines

This exercise reinforced a core system administration principle : prioritize automation, documentation, and the "why" behind configuration choices over manual, one-off execution — a mindset directly transferable to enterprise identity and access management (IAM) and infrastructure security roles.

